BOOLEAN ALGEBRA

Circuits and Gates

A logic circuit is a type of electrical circuit widely used in computers and other electronics devices. The simplest logic circuit called gates have the following properties:

1. Current flows to the circuit through one or two connectors called input lines and from the circuit through a connector called output lines.
2. The current in any of the input or output lines may have either of the two possible voltage levels. The higher is denoted by 1 and the Lower is denoted by 0.

1 is sometimes referred to as True or ON and 0 is referred to as False or OFF.

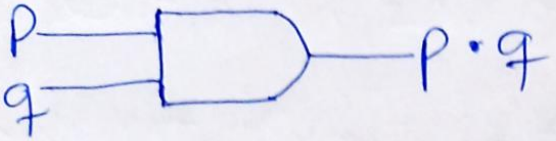
3. The voltage level of the output line depends on the voltage level(s) of the input line(s) according to the rules of logic.

The three (3) most basic types of gates are the:

1. AND gate
2. OR gate and
3. NOT gate

1. The AND gate

The AND gate is an electronic device which receives input 1, when both inputs are 1 and the output is 1.

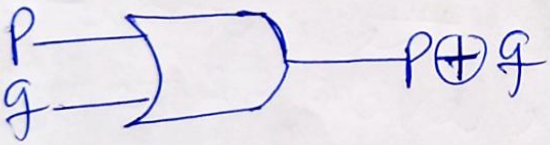


Truth Table

p	q	p·q	p⊕q
1	1	1	1
1	0	0	1
0	1	0	1
0	0	0	0

② The OR gate

The OR gate receives two inputs each in the form of a binary digit ie 0 or 1 and produces a binary digit as output depending upon the values of the two inputs OR gate.



p	q	p⊕q
1	1	1
1	0	1
0	1	1
0	0	0

③ The NOT gate

The NOT gate is sometimes called the INVERTOR gate. The invertor is a device ~~which~~ with one input and one output and has the following symbols.

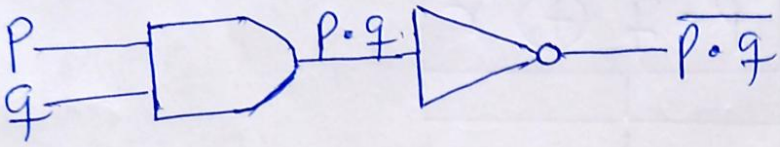
NOT gate

P	q	$\overline{P}$
1	1	0
1	0	0
0	1	1
0	0	1

P	$\overline{P}$
1	0
1	0
0	1
0	1

④ NAND gate (NOT AND gate)

The NAND gate is logically equivalent to a NOT gate in series with an AND gate.

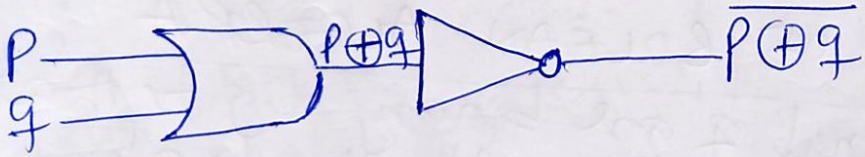


Truth table

P	q	$P \cdot q$	$\overline{P \cdot q}$
1	1	1	0
1	0	0	1
0	1	0	1
0	0	0	1

⑤ NOR gate

The NOR gate is logically equivalent to the NOT gate in series with the OR gate.



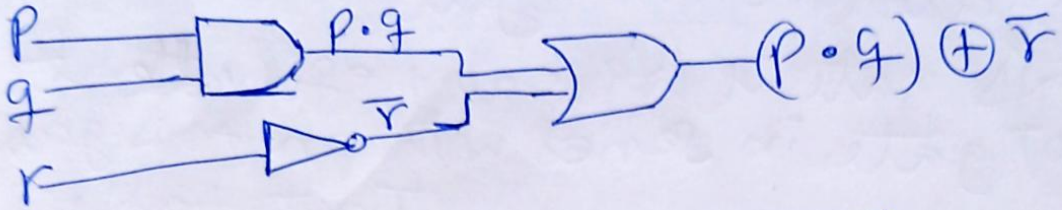
P	q	$P \oplus q$	$\overline{P \oplus q}$
1	1	1	0
1	0	1	0
0	1	1	0
0	0	0	1

Boolean sum = + = OR

Boolean product = • = AND

Que 1 (J)

Construct a logic circuit whose inputs are p, q, r and whose output is ~~p⊕q~~ $(\overline{p \cdot q}) \oplus \bar{r}$. When is the output equal to 1? Draw a truth Table.



p	q	p·q	r	r̄	p·q ⊕ r̄
1	1	1	1	0	1
1	1	1	0	1	1
1	0	0	1	0	0
1	0	0	0	1	1
0	1	0	1	0	0
0	1	0	0	1	1
0	0	0	1	0	0
0	0	0	0	1	1

PROPERTIES OF BOLEAN ALGEBRA

Suppose p and q are binary digits that is 1 or 0. These together with the logical connectives • and ⊕ and also the complement NOT ($\bar{p}$) form what is known as BOOLEAN ALGEBRA. The quantities p and q are known as Boolean variables. Expressions like p·q, p⊕q and $\bar{p}$ are known as Boolean expressions.

~~Pro~~ PROPERTIES (K)

For any Boolean variables p, q, r

① Idempotent properties

$p \cdot p = p$, $p \oplus p = p$

② Associative Property

$(p \cdot q) \cdot r = p \cdot (q \cdot r)$ $(p \oplus q) \oplus r = p \oplus (q \oplus r)$

③ Commutative

$p \oplus q = q \oplus p$ $p \cdot q = q \cdot p$

④ Distributive

$p \oplus (q \cdot r) = (p \oplus q) \cdot (p \oplus r)$

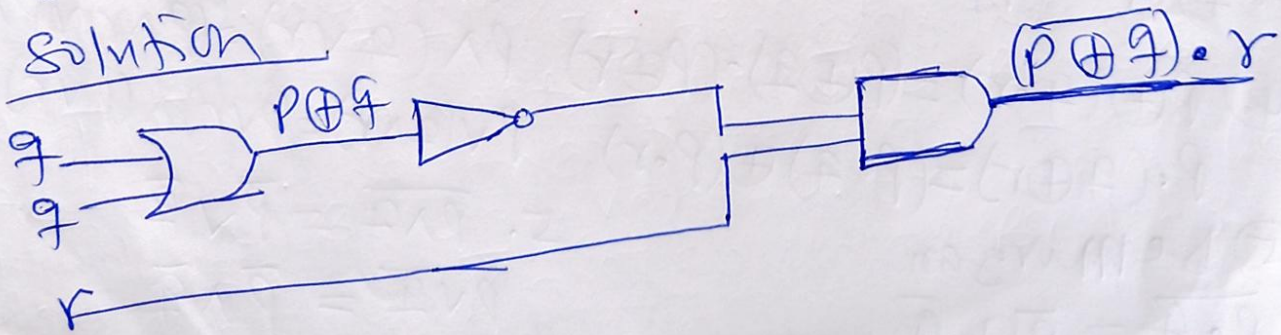
$p \cdot (q \oplus r) = (p \cdot r) \oplus (p \cdot q)$

⑤ De Morgan: $\overline{(p \cdot q)} = \overline{p} \oplus \overline{q}$ $\overline{p \oplus q} = \overline{p} \cdot \overline{q}$

⑥ Absorption: $p \oplus (p \cdot q) = p$ $p \cdot (p \oplus q) = p$

Qme2 Draw a circuit diagram for $(\overline{p \oplus q}) \cdot r$

Solution



p	q	r	$p \oplus q$	$\overline{p \oplus q}$	$(\overline{p \oplus q}) \cdot r$
1	1	1	1	0	0
1	1	0	1	0	0
1	0	1	1	0	0
1	0	0	1	0	0
0	1	1	1	0	0
0	1	0	1	0	0
0	0	1	0	1	1
0	0	0	0	1	0

Boolean Laws and Logic Laws are Logically equivalent

Boolean Laws

① Idempotent

$p \oplus p = p$, $p \cdot p = p$

② Commutative

$p \cdot q = q \cdot p$, $p \oplus q = q \oplus p$

③ Associative

$(p \oplus q) \oplus r = p \oplus (q \oplus r)$

$(p \cdot q) \cdot r = p \cdot (q \cdot r)$

④ $p \oplus (q \cdot r) = (p \oplus q) \cdot (p \oplus r)$

$p \cdot (q \oplus r) = (p \cdot q) \oplus (p \cdot r)$

⑤ Demorgan

$\overline{p \cdot q} = \overline{p} \oplus \overline{q}$

$\overline{p \oplus q} = \overline{p} \cdot \overline{q}$

Logic Laws

1. $p \wedge p = p$, $p \vee p = p$

② $p \wedge q = q \wedge p$

$p \vee q = q \vee p$

③ $(p \wedge q) \wedge r = p \wedge (q \wedge r)$

$(p \vee q) \vee r = p \vee (q \vee r)$

4. $p \wedge (q \vee r) = (p \wedge r) \vee (p \wedge q)$

$p \vee (q \wedge r) = (p \vee r) \wedge (p \vee r)$

5. $\overline{p \wedge q} = \overline{p} \vee \overline{q}$

$\overline{p \vee q} = \overline{p} \wedge \overline{q}$

(6) Absorption

$p \oplus (p \cdot q) = p$

$p \cdot (p \oplus q) = p$

6. $p \wedge (p \vee q) = p$

$p \vee (p \wedge q) = p$

(M)

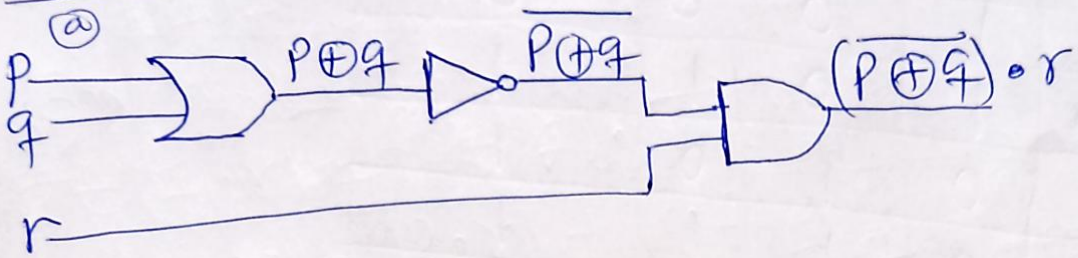
Que 3

Draw the circuit diagram ~~for~~ of:

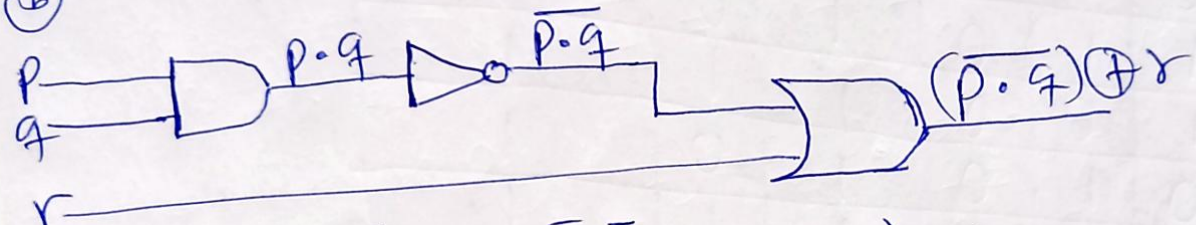
(a) $(\overline{p \oplus q}) \cdot r$ (b) $(\overline{p \cdot q}) \oplus r$ (c) $(\overline{p} \cdot \overline{q}) \cdot r$

Solution

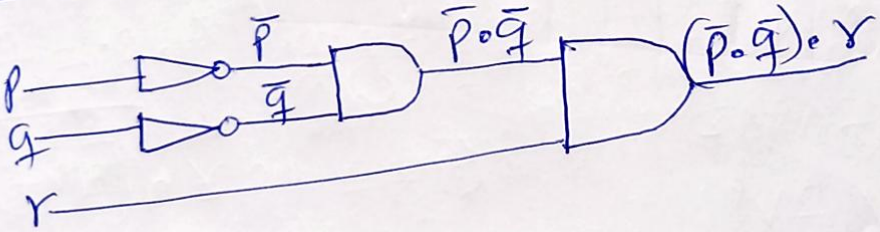
(a)



(b)



(c)



TRY QUESTIONS

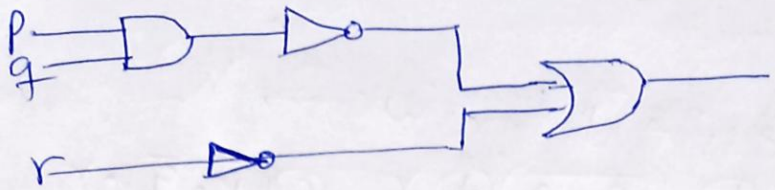
Draw the circuit diagrams of ~~the flow~~ the following

(a) $(\overline{p \cdot q}) \cdot r$ (b) $(p \oplus q) \cdot \overline{r}$

(c) $(p \oplus q) \cdot \overline{r} = (p \cdot \overline{r}) \oplus (q \cdot \overline{r})$

Que

Find the Boolean expression and Truth table for the electronic circuit shown below



Solution

p	q	r	$p \cdot q$	$\overline{p \cdot q}$	$\overline{r}$	$\overline{p \cdot q} \oplus \overline{r}$
1	1	1	1	0	0	0
1	1	0	1	0	1	1
1	0	1	0	1	0	1
1	0	0	0	1	1	1
0	1	1	0	1	0	1
0	1	0	0	1	1	1
0	0	1	0	1	0	1
0	0	0	0	1	1	1